

JAVA LAB MANUAL - TEST CASES

PROGRAM 1 — Hello World & Data Types

Test Case ID	Input	Expected Output
TC1	Run Program	Hello World
TC2	Byte.SIZE	8 bits
TC3	Character.SIZE	16 bits

PROGRAM 2 — Static, Local & Global Variables

Test Case ID	Variable Type	Expected Output
TC1	Static	10
TC2	Instance	20
TC3	Local	30

PROGRAM 3 — String Operations

Test Case ID	Operation	Expected Output
TC1	Length	5
TC2	Concatenate	Hello World
TC3	Substring	ell

PROGRAM 4 — Maximum of Three Numbers

Test Case ID	Input	Expected Output
TC1	10,25,15	25
TC2	5,5,5	5
TC3	-1,-5,-2	-1

PROGRAM 5 — Odd or Even

Test Case ID	Input	Expected Output
TC1	7	Odd
TC2	10	Even
TC3	0	Even

PROGRAM 6 — Constructors

Test Case ID	Constructor	Expected Output
TC1	Default	Number=0
TC2	Parameterized	Number=10
TC3	Name	John

PROGRAM 7 — Array of Objects

Test Case ID	Input	Expected Output
TC1	Alice,20	Displayed correctly
TC2	Bob,22	Displayed correctly
TC3	Charlie,21	Displayed correctly

PROGRAM 8 — Single Inheritance

Test Case ID	Method	Expected Output
TC1	show()	Parent class
TC2	display()	Child class
TC3	Inherited method	Accessible

PROGRAM 9 — Multiple Inheritance Using Interface

Test Case ID	Method	Expected Output
TC1	methodA()	Interface A output
TC2	methodB()	Interface B output
TC3	Implements	Both interfaces

PROGRAM 10 — Applet Life Cycle

Test Case ID	Method	Expected Output
TC1	init()	Applet Initialized
TC2	start()	Applet Started
TC3	paint()	Painting Applet

PROGRAM 11 — Division by Zero Exception

Test Case ID	Input	Expected Output
TC1	10/0	ArithmeticException
TC2	Handled exception	User-friendly message
TC3	Valid division	Correct result

PROGRAM 12 — Method Overloading

Test Case ID	Method	Expected Output
TC1	add(int,int)	10
TC2	add(float,float)	8.0
TC3	add()	15

PROGRAM 13 — Run-Time Polymorphism

Test Case ID	Object	Expected Output
TC1	Dog	Dog barks
TC2	Cat	Cat meows
TC3	Animal reference	Runtime decision

PROGRAM 14 — Negative Array Size Exception

Test Case ID	Array Size	Expected Output
TC1	-5	NegativeArraySizeException
TC2	Handled	Error message shown
TC3	Positive size	Array created

PROGRAM 15 — Null Pointer Exception

Test Case ID	Input	Expected Output
TC1	null string	NullPointerException
TC2	finally block	Executed always
TC3	Valid string	Length displayed

PROGRAM 16 — super Keyword

Test Case ID	Operation	Expected Output
TC1	super.display()	Parent display

TC2	super.num	100
TC3	child num	200

PROGRAM 17 — final Keyword

Test Case ID	Operation	Expected Output
TC1	Display constant	10
TC2	Modify constant	Compile-time error
TC3	Read final variable	Allowed

PROGRAM 18 — Constructor Overloading

Test Case ID	Constructor	Expected Output
TC1	Default	Unknown,0
TC2	Parameterized	John,20
TC3	Display	Correct values

PROGRAM 19 — Multilevel Inheritance

Test Case ID	Class	Expected Output
TC1	Grandparent	Grandparent class
TC2	Parent	Parent class
TC3	Child	Child class

PROGRAM 20 — Packages

Test Case ID	Operation	Expected Output
TC1	Import package	Successful
TC2	Create object	Object created
TC3	show()	Hello from package

Additional Viva Questions

1. Difference between class and object
2. What is inheritance in Java?
3. Explain polymorphism
4. Difference between constructor and method
5. What is method overloading?
6. What is exception handling?
7. Explain finally block
8. What is the use of super keyword?
9. What is the purpose of final keyword?
10. Explain packages in Java